



JKBOSE 2026

CLASS 11TH

SEQUENCES AND SERIES

MATHS

LECTURE-06

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TOPICS TO BE COVERED

1

Miscellaneous Exercise

A large, 3D rectangular frame with a red-to-orange gradient and a metallic sheen. It has a dashed pink border and contains the title text.

**MISCELLANEOUS
EXERCISE ON
CHAPTER – 8**

QUESTION

If f is a function satisfying $f(x+y) = f(x)f(y)$ for all $x, y \in \mathbb{N}$ such that $f(1) = 3$ and

$$\sum_{x=1}^n f(x) = 120, \text{ find the value of } n.$$

(Sol) Given that $\sum_{x=1}^n f(x) = 120$ & $f(1) = 3$.

$$\Rightarrow f(1) + f(2) + f(3) + \dots + f(n) = 120 \quad \text{--- (1)}$$

$$\text{Here } f(2) = f(1+1) = f(1) \times f(1) = 3^2$$

$$f(3) = f(2+1) = f(2) \times f(1) = 3^3$$

$$f(4) = f(3+1) = f(3) \times f(1) = 3^4$$

& so on $f(n) = 3^n$.

Using these values in (1), we get;

$$3 + 3^2 + 3^3 + 3^4 + \dots + 3^n = 120$$

$$\Rightarrow \frac{3(3^n - 1)}{3 - 1} = 120$$

$$\Rightarrow \cancel{3}(3^n - 1) = \cancel{120} \times 2$$

$$\Rightarrow 3^n - 1 = 80$$

$$\Rightarrow 3^n = 81$$

$$\Rightarrow 3^n = 3^4$$

$$\Rightarrow n = 4$$

QUESTION

The sum of some terms of G.P. is 315 whose first term and the common ratio are 5 and 2, respectively. Find the last term and the number of terms.

$$S_n = 315.$$

H/w. $\frac{a(r^n - 1)}{r - 1} = 315$

$$\frac{5(2^n - 1)}{2 - 1} = 315$$

$$n = ??$$

$$\text{Hence } a_n = ??$$

QUESTION

The first term of a G.P. is 1. The sum of the third term and fifth term is 90. Find the common ratio of G.P.

(Sol) Given that $a = 1$

$$ar^2 + ar^4 = 90.$$

$$\Rightarrow r^2 + r^4 - 90 = 0$$

$$\Rightarrow r^4 + r^2 - 90 = 0$$

$$\Rightarrow r^4 + 10r^2 - 9r^2 - 90 = 0$$

$$\Rightarrow r^2(r^2 + 10) - 9(r^2 + 10) = 0$$

$$\Rightarrow (r^2 - 9)(r^2 + 10) = 0.$$

$$\Rightarrow r^2 = 9, -10^{\times}$$

$$\Rightarrow r^2 = 9$$

$$\Rightarrow r = \pm 3$$

Therefore the required Common ratio = ± 3 .

QUESTION

$$a-1, b-7, c-21.$$



The sum of three numbers in G.P. is 56. If we subtract 1, 7, 21 from these numbers in that order, we obtain an arithmetic progression. Find the numbers.

(Sol) let the given numbers a, b, c .

Then according to the question;

$$a = a, b = ar, c = ar^2$$

$$(b-7) - (a-1) = (c-21) - (b-7)$$

$$\Rightarrow ar - a - 6 = ar^2 - ar - 14$$

$$\Rightarrow ar^2 - 2ar + a = 8$$

$$\Rightarrow a(r^2 - 2r + 1) = 8 \quad \text{--- (1)}$$

$$\text{Also } a + b + c = 56$$

$$\Rightarrow a + ar + ar^2 = 56$$

$$\Rightarrow a(r^2 + r + 1) = 56 \quad \text{--- (2)}$$

Dividing (2) by (1), we get;

$$\frac{a(r^2 + r + 1)}{a(r^2 - 2r + 1)} = \frac{56}{8} = 7$$

$$\Rightarrow r^2 + r + 1 = 7r^2 - 14r + 7$$

$$\Rightarrow 6r^2 - 15r + 6 = 0$$

$$\Rightarrow 3(2r^2 - 5r + 2) = 0$$

$$\Rightarrow 2r^2 - 5r + 2 = 0$$

$$\Rightarrow 2r^2 - 4r - r + 2 = 0$$

$$\Rightarrow 2r(r-2) - 1(r-2) = 0$$

$$\Rightarrow (2r-1)(r-2) = 0$$

$$\Rightarrow r = \frac{1}{2}, 2$$

$$\text{So, } a\left(\frac{1}{4} - \cancel{1} + 1\right) = 8$$

$$\text{or } a\left(\cancel{4} - \cancel{4} + 1\right) = 8$$

$$\Rightarrow a = 32, 8.$$

$$\Rightarrow b = 16.$$

$$c = 8, 32$$

Therefore the required numbers are

$$8, 16, 32$$

Hence the result.

a	b	c
32	16	8 ✓
✓ 8	✓ 16	✓ 32 ✓

$$r = \frac{1}{2}, 2$$

$$a = a = 32, 8$$

$$b = ar = 16, 16$$

$$c = ar^2 = 8, 32$$

QUESTION

A G.P. consists of an even number of terms. If the sum of all the terms is 5 times the sum of terms occupying odd places, then find its common ratio.

$$\frac{a(r^{2n} - 1)}{r - 1} = 5 \frac{(r^{2n} - 1)}{r^2 - 1} \times a$$

$$\Rightarrow \frac{1}{r - 1} = \frac{5}{(r - 1)(r + 1)}$$

$$\Rightarrow r + 1 = 5$$

$$r = 4.$$

$$a, ar, ar^2, ar^3, ar^4, \dots, ar^{2n-1} \checkmark$$

$$ar, ar^3, ar^5, ar^7, \dots, ar^{2n-1} \checkmark$$

$$r^2.$$

QUESTION

(PYQ JkBOSE)



If $\frac{a+bx}{a-bx} = \frac{b+cx}{b-cx} = \frac{c+dx}{c-dx}$ ($x \neq 0$), then show that a, b, c and d are in G. P.

Proof: Given that $\frac{a+bx}{a-bx} = \frac{b+cx}{b-cx} = \frac{c+dx}{c-dx}$

Applying Componendo-dividendo, we get;

$$\frac{\cancel{a+bx} + \cancel{a-bx}}{\cancel{a+bx} - \cancel{a-bx}} = \frac{\cancel{b+cx} + \cancel{b-cx}}{\cancel{b+cx} - \cancel{b-cx}} = \frac{\cancel{c+dx} + \cancel{c-dx}}{\cancel{c+dx} - \cancel{c-dx}}$$

$$\Rightarrow \frac{\cancel{2a}}{\cancel{2bx}} = \frac{\cancel{2b}}{\cancel{2cx}} = \frac{\cancel{2c}}{\cancel{2dx}}$$

$$\Rightarrow \frac{a}{b} = \frac{b}{c} = \frac{c}{d}$$

$$\Rightarrow \frac{b}{a} = \frac{c}{b} = \frac{d}{c}$$

This shows a, b, c, d are in G.P.

Hence proved

QUESTION

Let S be the sum, P the product and R the sum of reciprocals of n terms in a G.P.
Prove that $P^2 R^n = S^n$.

HP/W

QUESTION (PYQ JKBOSE)

$$2 + 4 + 8 + 16 + 32 + \dots \rightarrow \infty.$$



If a, b, c, d are in G.P, prove that $(a^n + b^n), (b^n + c^n), (c^n + d^n)$ are in G.P.

Proof: Since a, b, c & d are in G.P., so

$$a = a$$

$$b = ar$$

$$c = ar^2$$

$$d = ar^3$$

Now Consider the following:

$$\frac{b^n + c^n}{a^n + b^n} = \frac{a^n r^n + a^n r^{2n}}{a^n + a^n r^n}$$

$$\Rightarrow \frac{b^n + c^n}{a^n + b^n} = \frac{\cancel{a^n} (r^n + r^{2n})}{\cancel{a^n} (1 + r^n)}$$

$$\Rightarrow \frac{b^n + c^n}{a^n + b^n} = \frac{r^n + r^{2n}}{1 + r^n} \quad \text{--- (1)}$$

$$\text{Also } \frac{c^n + d^n}{b^n + c^n} = \frac{a^n r^{2n} + a^n r^{3n}}{a^n r^n + a^n r^{2n}}$$

$$\frac{c^n + d^n}{b^n + c^n} = \frac{\cancel{a^n} r^n (r^n + r^{2n})}{\cancel{a^n} r^n (1 + r^n)}$$

$$\Rightarrow \frac{c^n + d^n}{b^n + c^n} = \frac{r^n + r^{2n}}{1 + r^n} \quad \text{--- (2)}$$

From (1) & (2), we get:

$$\frac{b^n + c^n}{a^n + b^n} = \frac{c^n + d^n}{b^n + c^n}$$

$\Rightarrow a^n + b^n, b^n + c^n, c^n + d^n$ are in G.P.
Hence proved.

QUESTION

$$(3+2\sqrt{2}) : 3$$

If a and b are the roots of $x^2 - 3x + p = 0$ and c, d are roots of $x^2 - 12x + q = 0$, where a, b, c, d form a G.P. Prove that $(q + p) : (q - p) = 17 : 15$.

HP/wi

QUESTION

The ratio of the A.M. and G.M. of two positive numbers a and b , is $m : n$.

Show that $a : b = (m + \sqrt{m^2 - n^2}) : (m - \sqrt{m^2 - n^2})$.

H/W.

QUESTION

Find the sum of the following series up to n terms:

(i) $5 + 55 + 555 + \dots$

(ii) $.6 + .66 + .666 + \dots$

HP/w: $\frac{6}{10} + \frac{66}{100} + \frac{666}{1000} + \dots \#$

QUESTION



Find the 20th term of the series $\overset{a_1}{\underline{2 \times 4}} + \overset{a_2}{\underline{4 \times 6}} + \overset{a_3}{\underline{6 \times 8}} + \dots + n \text{ terms..}$

$$a_1 = 2 \times 4 = 2 \times 1 \times 2(1+1)$$

$$a_2 = 4 \times 6 = 2 \times 2 \times 2(2+1)$$

$$a_3 = 6 \times 8 = 2 \times 3 \times 2(3+1)$$

⋮

$$a_n = 2n \times 2(n+1)$$

$$\text{So, } a_{20} = 2 \times 20 \times 2(20+1) \\ = 40 \times 42.$$

$$a_n = 2n(2n+2) \\ = 4n^2 + 4n$$

$$\sum a_n = 4 \sum n^2 + 4 \sum n.$$

QUESTION

A farmer buys a used tractor for Rs 12000. He pays Rs 6000 cash and agrees to pay the balance in annual installments of Rs 500 plus 12% interest on the unpaid amount. How much will the tractor cost him?

h/w

QUESTION

Shamshad Ali buys a scooter for Rs 22000. He pays Rs 4000 cash and agrees to pay the balance in annual instalment of Rs 1000 plus 10% interest on the unpaid amount. How much will the scooter cost him?

H/w

QUESTION

A person writes a letter to four of his friends. He asks each one of them to copy the letter and mail to four different persons with instruction that they move the chain similarly. Assuming that the chain is not broken and that it costs 50 paise to mail one letter. Find the amount spent on the postage when 8th set of letter is mailed.

#.

QUESTION

A man deposited Rs 10000 in a bank at the rate of 5% simple interest annually. Find the amount in 15th year since he deposited the amount and also calculate the total amount after 20 years.

H/W

QUESTION

A manufacturer reckons that the value of a machine, which costs him Rs. 15625, will depreciate each year by 20%. Find the estimated value at the end of 5 years.

H/w.

QUESTION

150 workers were engaged to finish a job in a certain number of days. 4 workers dropped out on second day, 4 more workers dropped out on third day and so on. It took 8 more days to finish the work. Find the number of days in which the work was completed.

(15.)

(Sol) let the required number of days = n .

Then according to the question;

$$150 + 146 + 142 + \dots \text{upto } n \text{ terms} = 150(n-8)$$

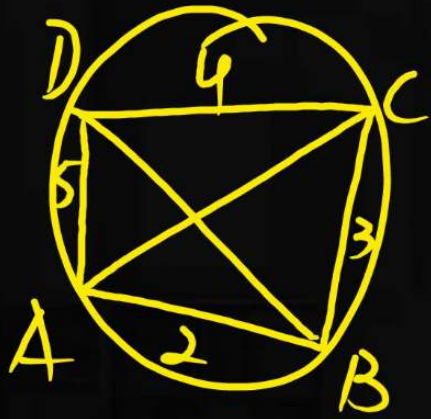
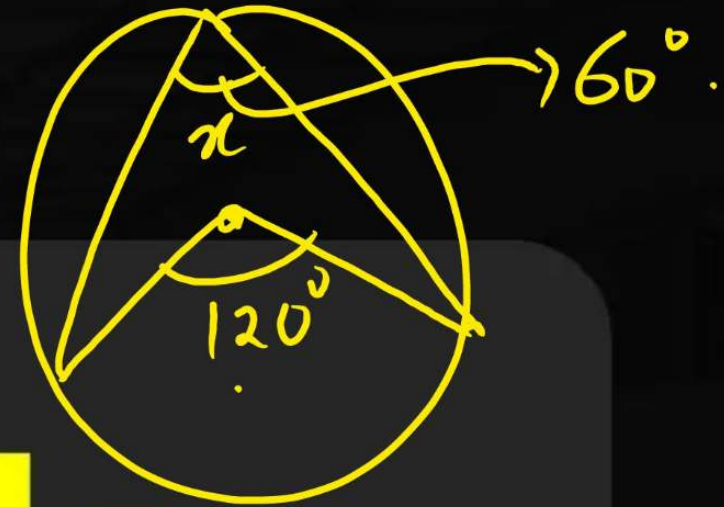
$$\Rightarrow \frac{n}{2} \{ 2 \times 150 + (n-1) \times -4 \} = 150n - 150 \times 8$$

$$\Rightarrow n(150 - 2n + 2) = 150n - 150 \times 8$$

$$\begin{aligned} & \Rightarrow \cancel{150n} - 2n^2 + 2n = \cancel{150n} - 150 \times 8 \\ & \Rightarrow \cancel{2}(-n^2 + n) = \cancel{-2}(150 \times 4) \\ & \Rightarrow -n^2 + n + 600 = 0 \\ & \Rightarrow n^2 - n - 600 = 0 \\ & \Rightarrow (n+24)(n-25) = 0 \\ & \Rightarrow n = 24, 25 \\ & \Rightarrow n = 25 \end{aligned}$$

✓ Set & functions $\swarrow \searrow$ (23) Marks.

✓ Algebra $\swarrow \searrow$ (25)



$AC \times BD$

$=$

(This chapter ends)

Thank You

Ptolemy's Theorem.